Shriman Oppilamani

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Education

University of California, Los Angeles Expected June 2028
Master of Science in Computer Science | Coursework: Computer Architecture, Computer Networks

Arizona State University May 2026
Bachelor of Science in Computer Science, GPA 4.10/4.00 | Dean's List

Skills

Languages: Rust, C, C++, Python, x86_64 Assembly, Java, SQL, Bash
Systems & Kernels: OS Dev (`no_std`), 4-Level Paging, IDT/TSS, DMA, PCI Bus Enumeration, POSIX, Systemd
Developer Tools: Wireshark, Valgrind, QEMU, GDB, Arch Linux, Make, Cargo, Git, MySQL, FastAPI, React

Projects

Melody OS (Bare-Metal Audio Operating System) Jan 2026 - Present
Rust (`no_std`), x86_64, Intel HDA, AHCI/SATA, smoltcp, QEMU

- Built a bare-metal x86_64 kernel from scratch in Rust (`no_std`): 4-level paging, heap allocator, IDT with TSS/IST stack-overflow protection, PIC timer and PS/2 drivers.
- Wrote PCI, AHCI SATA, and Intel HDA drivers; programmed codec widgets via verbs and streamed WAV audio from FAT32 over double-buffered DMA to S/PDIF optical output.
- Implemented an e1000 NIC driver with `smoltcp` and an MPD server on TCP port 6600, controllable from standard MPD clients.
- Tested via headless QEMU suites (heap, interrupts, WAV, FAT32, MPD); ran on Intel Core 2 Quad with glitch-free playback for 8 hours.

Low-Level POSIX Text Editor Nov 2025 - May 2026
C, POSIX APIs, Termios, Make, Unix

- Built a dependency-free C editor (1400 lines) on POSIX APIs and termios raw mode, with search, syntax highlighting, auto indentation.
- Engineered an ANSI-escape TUI rendering pipeline (no ncurses) with dynamic row-buffer batching to minimize screen redraw latency.
- Managed dynamic heap memory for unbounded row buffers and file I/O, verifying zero memory leaks via Valgrind.

Linux Kernel & Hardware Optimization Dec 2024 - Present
Linux Kernel, C, ACPI, Systemd, Arch Linux

- Diagnosed a CPU stuck at 200 MHz by tracing ACPI thermal polling and patched management routines to override broken EC polling, restoring 3.50 GHz sustained under load.
- Traced and patched proprietary Wi-Fi driver handshakes within `wpa_supplicant`, resolving persistent 802.1X EAP authentication failures.
- Tuned power-management states and systemd units to stabilize daily use on hardware without native Linux support.

Custom Pseudocode Compiler Frontend April 2025 - May 2025
C++, Lexical Analysis, Recursive Descent Parsing

- Architected a recursive descent compiler frontend in C++, transforming lexical token streams into a strongly-typed Abstract Syntax Tree (AST).
- Designed a memory-managed execution engine to optimize recursive AST evaluation for variable scopes, iterative loops, and nested control flow.

Experience

Software Engineering Capstone Developer Aug 2025 - May 2026
Varsity Sports Show | Python, Linux, Bash, MySQL

- Built a Flask backend that streams large media uploads in chunks directly to Backblaze B2, keeping memory use flat (15 GB files in under 500 MB RAM).
- Built REST APIs with session validation for gated subscriber access, backed by MySQL.
- Automated local container testing environments and deployment scripts with Bash, improving team velocity and delivery timelines.
- Delivered a production system for a commercial client in a team of 5.